



Causal Tongue-Tie

LLMs Can Encode Causal Direction
Yet Fail to Express It in Yes/No Answers

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MAIN · 599



arXiv



Code



WeChat

Wrong Yes/No. Recoverable causal direction.

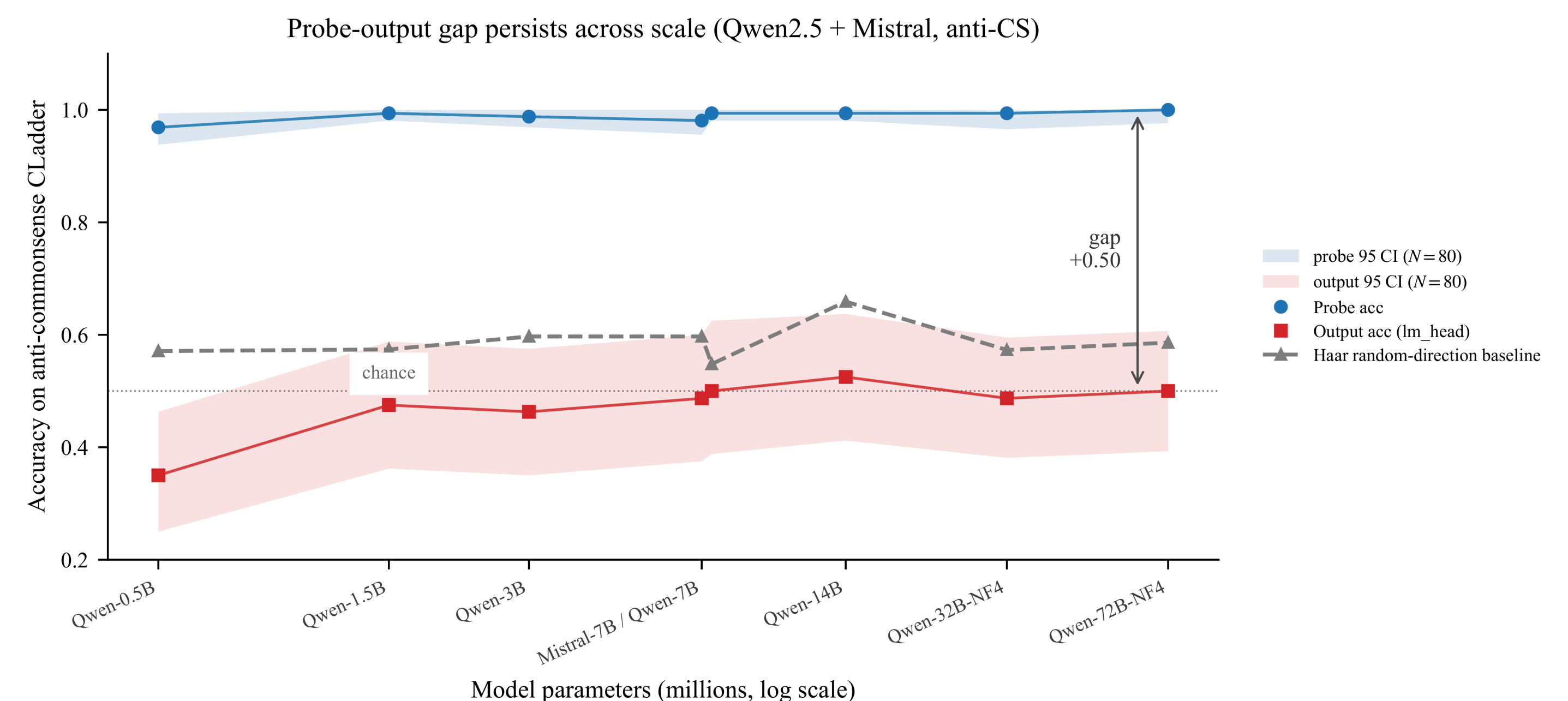
96.9–100%

Hidden-state probe accuracy

35.0–52.5%

Ordinary Yes/No accuracy

Eight models · 80 CLadder items
Evidence contradicts commonsense.



Qwen2.5 (0.5B–72B) + Mistral-7B; 95% CIs. Probe: 160 paired records; output: 80 questions. Reported layers; most selected by transfer accuracy. Fixed-layer checks retain the gap. Paper Fig. 3.

The missing distinction

Causal Parrots [1]: correct answers may follow surface associations.

Tongue-Tie: wrong answers can hide a readable direction.

	Causal answer <i>matches</i> evidence	Causal answer <i>contradicts</i> evidence
Probe recovers the causal direction	(a) Aligned correct. Both readouts pick the evidence-supported answer.	(b) Causal Tongue-Tie (ours). Probe recovers the causal direction; model gives the wrong Yes/No.
Probe does not recover the causal direction	(c) CAUSAL PARROT risk. Correct output may follow surface associations.	(d) Both readouts fail. Neither readout recovers the evidence-supported answer.

Adapted from paper Fig. 1. Probe failure does not prove information absence.

Two readouts, one frozen LLM

Given prompt x_i and gold answer y_i , compare a probe on hidden state $\mathbf{h}_i^{(\ell)}$ with the model's Yes/No.

$$g_{\mathbf{w}}(\mathbf{h}_i^{(\ell)}) = \sigma(\mathbf{w}^\top \mathbf{h}_i^{(\ell)} + b)$$

FIT PROBE	→	FREEZE + TEST
Commonsense evidence		Conflicting evidence

Last prompt token; no LLM weight updates.

01 Is the signal learned?

Pretrained weights Random weights
99.4% **50.0%**

Qwen-7B probe accuracy; same pipeline, two random seeds.

02 Does it affect computation?

KL change · target erasure KL change · random control
–96.3% **+0.7%**

Qwen-7B, layer 27, 30 commonsense pairs. SVD direction, not probe weights. KL change, not accuracy gain.

Scalar swaps did not reliably flip answers in this test.

03 Can it be expressed?

Contrast the two directions.

Changing Yes/No to A/B alone is not enough.

Accuracy	Proposition + verdict	Proposition + choice	Edge + verdict	Edge + choice
Qwen-7B	.57	.49	.58	.95
Qwen-14B	.74	.70	.75	1.00
Mistral-7B	.76	.53	.96	.97
Llama-3.1-8B	.50	.42	.51	.96

Separate 38-item test, 76 queries/condition. Rounded values, paper Table 3. Mistral also succeeds with edge verdicts.

Takeaway

Separate internal recoverability
from external expression.

Controlled textual tasks; template/layer dependent. Not human-like understanding or a universal repair.

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[1] Zečević et al. (2023). Causal Parrots: Large Language Models May Talk Causality But Are Not Causal. TMLR.

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